

A084 · SigRepo: A Platform For Storing, Sharing, and Comparing Signatures

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High-throughput studies generate a growing volume of omics signatures, the ranked or thresholded feature sets derived from differential analysis. Most remain locked in supplementary tables with inconsistent metadata and no shared representation, so signatures are rarely reused, and comparing a new result against prior work remains largely manual. We present SigRepo, an open-source platform for storing, sharing, and analyzing omics signatures and collections.

SigRepo standardizes signatures using OmicSignature, an R6 object specification built in-house by our lab and released under GPL-3, which pairs a curated feature set with its underlying differential-expression table and a controlled metadata vocabulary (organism, phenotype, sample type, platform, assay type). The platform comprises a MySQL database, a Plumber REST API, an R client package, and a web interface, distributed as containerized services deployable locally or in the cloud. The repository spans transcriptomic, proteomic, metabolomic, and genetic-variant assays, organized into signature collections with per-user access control and selective sharing, and signatures are currently being ingested on an ongoing basis.

Beyond storage, SigRepo treats stored signatures as analyzable objects. Any set of signatures can be compared by feature overlap (Jaccard index with Fisher exact tests), by rank-based Kolmogorov-Smirnov statistics, or by gene set enrichment analysis, with results rendered as interactive similarity heatmaps and, for GSEA, per-pair leading-edge plots. Enrichment is provided through hypeR for gene-based signatures and through hypeR-GEM for metabolomics signatures, so every assay type represented in the repository is supported. Because analyses are exposed as discrete endpoints over a common signature representation, external resources and analysis engines can be added without schema changes, letting the platform grow as community resources appear.

Signatures are most valuable when they can move between tools. Because SigRepo exposes its holdings through a documented REST API over a common object specification, it can serve as a bridge to established signature-based software; a Model Context Protocol (MCP) server extends the same access to language-model agents, which can search the repository and run comparisons and enrichment against stored data. We are working toward interoperation with tools such as signatureSearch, so that signatures held in SigRepo can be carried into connectivity-mapping and drug-repurposing analyses and their results

returned to the repository. The same interfaces allow new external gene-set and signature resources to be attached as they emerge, positioning SigRepo as a hub through which signatures move between analyses rather than a static archive.

SigRepo: An Integrated Platform for Storing, Sharing, and Comparing Omics Signatures

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Implementation

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Analysis in place

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Interoperability

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Availability. SigRepo is freely available at <https://sigrepo.org>, with source code at github.com/montilab/SigRepo and github.com/montilab/SigRepo_Server.

References. [1] Federico A, Monti S. hypeR: an R package for geneset enrichment workflows. *Bioinformatics* 2020. [2] Liberzon A, et al. The Molecular Signatures Database hallmark gene set collection. *Cell Syst* 2015.